

Exact endpoint windows: Gamma coefficients, paired Schur volumes, and intrinsic sensitivity

13 September 2026

This continuation evaluates the original endpoint quantity of the complete written criterion at source revision `c720f40530eed2f5969dbabe94dfd3fddc0f507f` of PR 23. The source criterion and the supplied *Endpoint degree selection in the actual gamma-corrected coordinates* were read completely, together with GC.1–59 and GMT.1–42, including GMT.29a–b. The present work proves the arbitrary-window Schur map, its exact determinant cancellations, a finite rational interval certificate, and a new sensitivity estimate through the actual angles between canonical least-lift spaces. The finite input list grows through degree $2(n + r)$. Every statement below is finite. No estimate for that growing list or its arithmetic conditioning is presumed.

1 The original source and the complete finite input map

Let $h(s)$ be the given monic packet of actual zeros, with their complete orders, of

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s).$$

Retain its entire quotient $v_h = g/h$ and original positive density

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}. \quad (\text{EW.1})$$

The complete analytic construction in GC.2–9 proves that w_h has every polynomial moment and exponential moments of every order less than $\pi/2$. Its mechanism is the theta seed $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, the equality $\mathcal{M}\Theta\phi_0 = g$, contour rotation through every angle $|\beta| < \pi/4$, and exact local division by h . In particular $|v_h(\sigma + it)|$ is bounded, uniformly for σ in each fixed compact real interval, by $C_{j,\beta,h}(1+|t|)^{-j}e^{-\beta|t|}$ for every $j \geq 0$ and $0 < \beta < \pi/4$. These proofs and their original constants are retained in the accompanying complete GC source. The zero set of $v_h(1/2 + it)$ is discrete, so $w_h > 0$ almost everywhere. Its convolution has positive mass μ_h^k .

Throughout this paper

$$k \geq 1, \quad c = k/2, \quad S = s_1 + \cdots + s_k = c + iu, \quad u = t_1 + \cdots + t_k, \quad (\text{EW.2})$$

$$\langle P, Q \rangle_H = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) m_{h,k}(u) du.$$

The pairing is conjugate-linear in its first argument. The sum chart $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ has determinant one. Fubini therefore proves both the convolution in (EW.1) and the full mass μ_h^k , with no average coordinate or mass division. Every nonzero polynomial has finitely many real-line zeros, so its squared norm in (EW.2) is strictly positive.

Fix the original monic relation $\chi \in \mathbb{C}[S]$, of degree $q \geq 1$. For the packet source it generates the exact intersection $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$ under $S = \sum s_i$. Define $\mathcal{C} = \mathbb{C}[S]/(\chi)$ in the ordered coordinates $1, S, \dots, S^{q-1}$. The quotient map is the actual remainder map. Its injection into the tensor arithmetic jets is

$$\eta : \mathcal{C} \longrightarrow (\mathbb{C}[s]/(h))^{\otimes k}, \quad [P] \longmapsto [v_h]_h^{\otimes k} P(s_1 + \cdots + s_k). \quad (\text{EW.3})$$

Complete local cancellation makes the constant term of v_h nonzero at every selected centre. Each local truncated power series is invertible by recursively solving its product with an inverse to equal one. The Chinese remainder isomorphism then proves that $[v_h]_h$ is a unit. The unfactored polynomial map $\mathbb{C}[S] \rightarrow (\mathbb{C}[s]/(h))^{\otimes k}$ in (EW.3) consequently has kernel equal to the stated intersection ideal (χ) . Factoring by that ideal gives the displayed η with kernel zero, proving injectivity without discarding a jet.

For $N \geq q - 1$, put $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and $\mathcal{P}_{-1} = 0$. Let H_N be its source matrix in the original monomials. Let $J_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}^q$ be monic remainder and $B_N : \mathbb{C}^{N-q+1} \rightarrow \mathbb{C}^{N+1}$ be multiplication by χ . Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} \mathcal{C} \longrightarrow 0. \quad (\text{EW.4})$$

In particular B_N is injective, J_N is surjective, and $\ker J_N = \text{im } B_N$.

For later coefficient evaluation keep all Gamma constants:

$$\lambda > 0, \quad \alpha = 2\lambda, \quad \beta = k\alpha, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad C_k = c_\lambda^k. \quad (\text{EW.5})$$

Let the real monic polynomials $b_j^{(\lambda)}$ be specified by

$$\sum_{j \geq 0} \frac{b_j^{(\lambda)}(t)}{j!} z^j = (1 + z^2)^{-\lambda} e^{t \arctan z}, \quad |z| < 1, \quad b_{j+1} = t b_j - j(j + \alpha - 1) b_{j-1}, \quad (\text{EW.6})$$

where the logarithms defining the generating function are the branches from zero, $b_0 = 1$, and $b_1 = t$. Differentiation of the displayed generating function proves the recurrence and monicity. The beta integral and Fourier inversion give the literal convolution mass and Laplace transform

$$r_{\lambda,k} = r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}, \quad \int e^{\theta u} r_{\lambda,k}(u) du = C_k (\cos \theta)^{-\beta}, \quad |\Re \theta| < \pi/2. \quad (\text{EW.7})$$

Their complete beta-integral proof with the changes $t = 2v$ and $u = \sum t_i$ is GC.10–15. In particular the reference monic norms at $\theta = 0$ are $C_k j! (\beta)_j$. They also follow directly by integrating the product of two functions (EW.6), which gives $C_k (1 - zw)^{-\beta}$, and comparing its finite derivatives.

Define the given one-factor coefficients and their finite powers by

$$c_{h,j} = \frac{\int b_j^{(\lambda)}(t) w_h(t) dt}{c_\lambda j! (\alpha)_j}, \quad d_l = \sum_{j_1 + \dots + j_k = l} \prod_{a=1}^k (\alpha)_{j_a} c_{h,j_a}. \quad (\text{EW.8})$$

Multiplying the k generating functions (EW.6) and comparing the coefficient of z^l gives the finite polynomial identity

$$b_l^{(k\lambda)}(\sum t_i) = \sum_{j_1 + \dots + j_k = l} \frac{l!}{\prod j_i!} \prod_i b_{j_i}^{(\lambda)}(t_i).$$

Integrating every term against the full product measure proves

$$\int b_l^{(k\lambda)}(u) m_{h,k}(u) du = C_k l! d_l. \quad (\text{EW.9})$$

All sums here are finite and all their terms are integrable. For any specified polynomials $p_a \in \mathcal{P}_M$, perform monic polynomial subtraction in the actual u variable to find the unique coefficients $L_{ab,l}$ in

$$\overline{p_a(c + iu) p_b(c + iu)} = \sum_{l=0}^{2M} L_{ab,l} b_l^{(k\lambda)}(u).$$

The complete coefficient-to-Gram map is therefore

$$H(p)_{ab} = C_k \sum_{l=0}^{2M} l! L_{ab,l} d_l. \quad (\text{EW.10})$$

It is a degree- k real polynomial map from the actual coefficient vector to Hermitian forms. The precise tests contain $c = k/2$, the two phases $\pm i$, and, when a relation column is used, both $\chi(c + iu)$ and $\bar{\chi}(c - iu)$. In particular (EW.10) constructs H_M from $c_{h,0}, \dots, c_{h,2M}$ and the stated k, λ . At $l = 0$, $c_{h,0} = \mu_h/c_\lambda$, so the same map gives $H_{00} = \mu_h^k$.

2 Canonical lifts and one arbitrary-window block

For each admitted degree define the exact matrices and maps

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = H_N^{-1} J_N^* G_N : \mathbb{C}^q \longrightarrow \mathbb{C}^{N+1}, \quad V_N = \det G_N. \quad (\text{EW.11})$$

Positivity of H_N and surjectivity of J_N prove $K_N > 0$. Multiplication gives $J_N R_N = I$ and $B_N^* H_N R_N = 0$. For every $x \in \mathbb{C}^q$, any lift is $R_N x + B_N y$, and its norm is exactly $x^* G_N x + y^* B_N^* H_N B_N y$. This proves the unique minimum property and $R_N^* H_N R_N = G_N$.

Let E_N include the first q monomials. The square coefficient matrix $[E_N, B_N]$ has determinant one: its columns are ordered by successive degree and each leading coefficient is one. Subtracting boundary columns to replace E_N by R_N retains determinant one. Its Gram is block diagonal, so

$$V_N = \frac{\det H_N}{\det(B_N^* H_N B_N)}. \quad (\text{EW.12})$$

Empty determinants are one. This fixes the relation determinant and its index $N - q + 1$ in the original quotient coordinates.

Take integers

$$n \geq q, \quad r \geq 1, \quad m = n + r, \quad d = r + 1. \quad (\text{EW.13})$$

In $\mathcal{P}_m$ split the ordered monomials into degrees $0, \dots, n-1$ and the d appended columns $S^n, \dots, S^m$. Write

$$H_m = \begin{pmatrix} H_{n-1} & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_{n-1}^{-1} C_D, \quad J_m = [J_{n-1}, J_D], \quad F = J_D - J_{n-1} H_{n-1}^{-1} C_D. \quad (\text{EW.14})$$

The map

$$\Psi : \mathbb{C}^n \oplus \mathbb{C}^d \longrightarrow \mathbb{C}^{m+1}, \quad (x, y) \longmapsto (x - H_{n-1}^{-1} C_D y, y) \quad (\text{EW.15})$$

is invertible with determinant one. Direct multiplication gives $\Psi^* H_m \Psi = \text{diag}(H_{n-1}, W)$ and $J_m \Psi = [J_{n-1}, F]$. Thus $W > 0$; F maps the appended orthogonal residual polynomials to their original quotient classes. No quotient is independently chosen.

Let W_j be the first j rows and columns of W , F_j its first j quotient columns, and $Z_j = F_j^* G_{n-1} F_j$ for $1 \leq j \leq d$. The same map (EW.15), restricted to these columns, gives the exact positive-kernel update

$$K_{n+j-1} = K_{n-1} + F_j W_j^{-1} F_j^*. \quad (\text{EW.16})$$

Eliminating the two diagonal blocks of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ in the two orders proves $\det(I + AB) = \det(I + BA)$. Use this identity in (EW.16) to obtain

$$\boxed{\mathcal{R}_j := \frac{V_{n-1}}{V_{n+j-1}} = \frac{\det(W_j + Z_j)}{\det W_j}, \quad \mathcal{R}_0 = 1.} \quad (\text{EW.17})$$

All the determinants on the right are of size j , whereas its equivalent kernel calculation has size q . The matrices Z_j are positive semidefinite. Diagonalizing $W_j^{-1/2} Z_j W_j^{-1/2}$ proves $\mathcal{R}_j \geq 1$. Nested minimization also proves $G_{N+1} \preceq G_N$, so $\mathcal{R}_j$ is nondecreasing in j .

Let $p_N(S)$ be the unique monic polynomial of degree N orthogonal to $\mathcal{P}_{N-1}$, and let ω_N be its original squared norm. Existence and uniqueness follow by solving the positive lower-degree projection problem; its leading coefficient remains one. The first diagonal entry $w = W_1$

is ω_n . Eliminating the first r residual columns from the final one gives $\omega_m = \det W_d / \det W_r$. Consequently

$$U := \frac{\omega_m}{\omega_n} = \frac{\det W_d}{\det W_r w}, \quad \mathcal{R} := \frac{V_{n-1}V_n}{V_{m-1}V_m} = \frac{\mathcal{R}_r \mathcal{R}_d}{\mathcal{R}_1} \geq 1. \quad (\text{EW.18})$$

The last inequality also follows by multiplying $V_{n-1}/V_{m-1} \geq 1$ and $V_n/V_m \geq 1$. Both numerator and denominator norms carry their literal masses.

For the Gamma source (EW.7), the same J_m and χ give matrices $W^\Gamma, F^\Gamma, Z_j^\Gamma$ by the identical maps. For their original source determinants put $\mathfrak{D}_j = \det H_{j-1}$ for $j \geq 1$ and $\mathfrak{D}_0 = 1$. For the relation determinant on exactly j columns put $\mathfrak{B}_j = \det(B_{q+j-1}^* H_{q+j-1} B_{q+j-1})$ for $j \geq 1$ and $\mathfrak{B}_0 = 1$. The reference determinants use the same columns and carry the superscript Γ . Writing the supplied corrections as $X_j = \mathfrak{D}_j / \mathfrak{D}_j^\Gamma$, $Y_j = \mathfrak{B}_j / \mathfrak{B}_j^\Gamma$ and $T_N = X_{N+1} / Y_{N-q+1}$, the relation (EW.12) proves

$$U = (n+1)_r (n+\beta)_r \frac{X_{m+1} X_n}{X_m X_{n+1}}, \quad \mathcal{R} = \frac{V_{n-1}^\Gamma V_n^\Gamma}{V_{m-1}^\Gamma V_m^\Gamma} \frac{T_{n-1} T_n}{T_{m-1} T_m}. \quad (\text{EW.19})$$

Indeed $\omega_j = C_k j! (\beta)_j X_{j+1} / X_j$ follows by taking successive source determinants in the original monic basis. Taking the ratio gives the first line with its exact rising factorials. The new symbol $\mathcal{R}_j$ in (EW.17) has no effect on the original positive arithmetic correction T_N .

3 Endpoint roots and exact paired-volume cancellations

Put $a = 1/(2r)$. The original endpoint expression is

$$\mathcal{E}_{n,r} = U^a \sinh(a \log \mathcal{R}) = \frac{(U\mathcal{R})^a - (U/\mathcal{R})^a}{2}. \quad (\text{EW.20})$$

All roots are the unique positive real roots. The equality follows from the definition of the real hyperbolic sine and positivity. It evaluates the existing quantity, preserving its two original norm endpoints and all four original volume endpoints.

There is a further exact cancellation inside its two root arguments. Set

$$p = \det W_r, \quad s = \det W_d, \quad e = w + Z_1, \quad D = \det(W_r + Z_r) \det(W_d + Z_d).$$

The scalar Z_1 is real nonnegative, and all four quantities p, s, e, D are strictly positive. Substituting (EW.17)–(EW.18) without changing a coordinate gives

$$U = \frac{s}{pw}, \quad \mathcal{R} = \frac{Dw}{pse}, \quad \boxed{U\mathcal{R} = \frac{D}{p^2e}, \quad \frac{U}{\mathcal{R}} = \frac{s^2e}{w^2D}}. \quad (\text{EW.21})$$

Thus the common highest source determinant cancels from the first argument; the common prefix source determinant cancels from the second. The first and last original monic norms have not been set equal or removed from the underlying source. Formula (EW.21) is the exact rational map explaining these cancellations.

It is sometimes useful to retain the small difference in (EW.20) as one positive numerator. With $P = (U\mathcal{R})^a$ and $Q = (U/\mathcal{R})^a$, the finite factorization $P^{2r} - Q^{2r} = (P - Q) \sum_{j=0}^{2r-1} P^{2r-1-j} Q^j$ proves

$$\mathcal{E}_{n,r} = \frac{U(\mathcal{R}^2 - 1)}{2\mathcal{R} \sum_{j=0}^{2r-1} P^{2r-1-j} Q^j}. \quad (\text{EW.22})$$

The denominator remains positive at $\mathcal{R} = 1$, and the numerator is then exactly zero. This representation preserves the sign of the endpoint quantity and avoids subtracting two enclosures before their common positive factor has been identified.

Every entry in (EW.14) is determined by (EW.10) through degree $2m = 2(n+r)$. Each J_N and B_N is computed by monic division or multiplication by the same χ . Equations (EW.11)–(EW.22) therefore prove the complete finite map

$$\boxed{(c_{h,0}, \dots, c_{h,2(n+r)}; k, \lambda, \chi) \mapsto (H_m, J_m, W, F, G_{n-1}) \mapsto \mathcal{E}_{n,r}.} \quad (\text{EW.23})$$

It consists of stated polynomial operations, inverses of matrices whose positivity has been proved, determinants, and positive roots.

For reference the endpoint selection follows from the original degree-wise estimate by a finite argument. Write $x_j = \frac{1}{2} \log(V_{n+j-1}/V_{n+j+1}) \geq 0$ and $A_j = \sqrt{\omega_{n+j+1}/\omega_{n+j}}$. The local estimate is $0 \leq \epsilon_{n+j} \leq A_j \sinh x_j$. If an x_j is zero, the minimum allowance is zero. Otherwise $(\log \sinh x)'' = -1/\sinh^2 x < 0$ proves finite Jensen, hence $\prod_j \sinh x_j \leq \sinh(\sum_j x_j/r)^r$. The actual products give $\prod_j A_j = \sqrt{U}$ and $2 \sum_j x_j = \log \mathcal{R}$. Multiplying the local estimates and taking the positive r th root therefore proves

$$\min_{0 \leq j < r} \epsilon_{n+j} \leq \mathcal{E}_{n,r}. \quad (\text{EW.24})$$

The local arithmetic trace inequality and its retained loss terms are proved in the complete source TVB.32–34a; (EW.24) is their exact finite selection, not an added local estimate or an asymptotic claim.

4 Exact projection between the canonical representative spaces

The following result supplies more information than the separate determinant values. Use the common ambient space $(\mathcal{P}_m, \langle \cdot, \cdot \rangle_H)$, with its original source inner product. Let $\iota_j : \mathbb{C}^{j+1} \rightarrow \mathbb{C}^{m+1}$ be monomial inclusion. The least-lift space and its exact isometry are

$$L_j = \iota_j R_j \mathbb{C}^q \subset \mathcal{P}_m, \quad \mathcal{J}_j : (\mathbb{C}^q, G_j) \longrightarrow L_j, \quad x \longmapsto \iota_j R_j x. \quad (\text{EW.25})$$

Here $\mathcal{J}_j^* H_m \mathcal{J}_j = G_j$ proves the word isometry; its inverse on L_j is the original remainder map. The original arithmetic correction T_N retains its meaning in (EW.19).

Theorem 4.1 (Canonical projection and all its moving directions). *For $q-1 \leq a < b \leq m$, let $\Pi_b : \mathcal{P}_m \rightarrow \mathcal{P}_m$ be the ambient orthogonal projection with range L_b , including that range by its fixed polynomial inclusion. It obeys*

$$\boxed{\Pi_b \iota_a R_a x = \iota_b R_b x \quad (x \in \mathbb{C}^q).} \quad (\text{EW.26})$$

Its restriction $L_a \rightarrow L_b$ is invertible. Its squared singular values, with precisely the source inner products in (EW.25), are the eigenvalues σ_i^2 of

$$G_a^{-1/2} G_b G_a^{-1/2}, \quad 0 < \sigma_i \leq 1. \quad (\text{EW.27})$$

At most $s_{a,b} = \min(q, b-a)$ of these values are less than one, and

$$\begin{aligned} \prod_i \sigma_i^2 &= V_b/V_a, \\ \|\Pi_a - \Pi_b\|_1 &= 2 \sum_i \sqrt{1 - \sigma_i^2}, \\ \|\Pi_a - \Pi_b\|_1 &\leq \min\{2\sqrt{s_{a,b} \log(V_a/V_b)}, 2s_{a,b}\}. \end{aligned} \quad (\text{EW.28})$$

The trace norm is for operators in the stated ambient inner product.

Proof. Both representatives in (EW.26) have remainder x . Their difference belongs to $\chi\mathcal{P}_{b-q}$ and is therefore orthogonal to L_b by (EW.11). The second representative belongs to L_b ; this proves the orthogonal projection identity. The same identity and injectivity of each lift show that the restriction is invertible. In the quotient coordinates it is the identity map from metric G_a to metric G_b . The explicit maps from Euclidean coordinates are $u \mapsto \iota_a R_a G_a^{-1/2} u$ and $v \mapsto \iota_b R_b G_b^{-1/2} v$. Their matrices give the squared singular values (EW.27), and no original form or mass has been changed. A projection is a contraction, so these values lie in $(0, 1]$. Taking its determinant proves the first line of (EW.28).

The boundary space at degree b extends the one at degree a by exactly $b-a$ independent monic columns. Vectors in L_a are already orthogonal to the older boundary. Imposing orthogonality to these extra $b-a$ columns has rank at most $b-a$. Its kernel is $L_a \cap L_b$: any vector in that kernel is in $\mathcal{P}_b$ and orthogonal to its full boundary, hence is a minimum representative. The intersection therefore has dimension at least $q - (b-a)$ when this number is positive. A unit vector in L_a is unchanged in norm by Π_b precisely when it belongs to L_b , by orthogonal Pythagoras. Thus at most $s_{a,b}$ singular values are less than one.

For a complete trace-norm calculation choose orthonormal eigenvectors $x_i \in L_a$ for the squared restriction and put $y_i = \Pi_b x_i / \sigma_i \in L_b$. Orthogonality gives $\langle x_i, y_j \rangle = \sigma_j \delta_{ij}$ and the y_i form an orthonormal basis of L_b . If $\sigma_i = 1$, then $y_i = x_i$. If $\sigma_i < 1$, set $z_i = (y_i - \sigma_i x_i) / \sqrt{1 - \sigma_i^2}$. Directly taking all these inner products proves that the planes $\text{span}(x_i, z_i)$ are mutually orthogonal and orthogonal to the common directions. In such a plane the matrix of $\Pi_a - \Pi_b$ is exactly

$$\begin{pmatrix} 1 - \sigma_i^2 & -\sigma_i \sqrt{1 - \sigma_i^2} \\ -\sigma_i \sqrt{1 - \sigma_i^2} & -(1 - \sigma_i^2) \end{pmatrix}.$$

Its eigenvalues are $\pm \sqrt{1 - \sigma_i^2}$. On the common directions and the orthogonal complement of $L_a + L_b$ the difference is zero. This proves the exact trace-norm sum. Finally $1 - x \leq -\log x$ for $0 < x \leq 1$, because the derivative of $x - 1 - \log x$ is $1 - 1/x$ and its value at one is zero. Cauchy-Schwarz on the at most $s_{a,b}$ nonzero terms gives the first bound in (EW.28); bounding each square root by one gives the second bound independently. $\square$

5 Window sensitivity and exact cancellation of mass changes

The next calculation holds for every continuously differentiable family of positive source matrices $H_m(t)$, using all their principal submatrices and the fixed χ, J_j . Thus it applies both to the actual coefficient forms and to a segment between their certified finite enclosures. Write $E = \dot{H}_m$ and define the real spectral spread

$$\mathfrak{o}_H(E) = \lambda_{\max}(H_m^{-1/2} E H_m^{-1/2}) - \lambda_{\min}(H_m^{-1/2} E H_m^{-1/2}) \geq 0. \quad (\text{EW.29})$$

The two congruences in this expression specify its exact coordinate map; the source H_m remains in every evaluation.

Embed the coefficient vector of p_j in $\mathcal{P}_m$ and set

$$P_j = \frac{\iota_j p_j p_j^*}{\omega_j}, \quad Q_j = \iota_j R_j G_j^{-1} R_j^* \iota_j^*. \quad (\text{EW.30})$$

These are contravariant matrices; their products $P_j H_m$ and $Q_j H_m$ are respectively the ambient orthogonal projections onto the monic orthogonal line and L_j . To verify this, multiply each by itself and use $p_j^* H_j p_j = \omega_j$ or $R_j^* H_j R_j = G_j$. The resulting operators are idempotent and self-adjoint in H_m and have the stated ranges. The matrices $H_m^{1/2} P_j H_m^{1/2}$ and $H_m^{1/2} Q_j H_m^{1/2}$ are their exact Euclidean isometric conjugates.

Differentiating the norm of the monic polynomial gives $\dot{\omega}_j = p_j^* \dot{H}_j p_j$: its derivative has degree less than j , and both cross terms vanish by its original orthogonality. Differentiating $J_j R_j = I$ makes $\dot{R}_j$ a boundary, so the same expansion gives $\dot{G}_j = R_j^* \dot{H}_j R_j$. The determinant derivative follows by expanding the determinant multilinearly in its columns, or by differentiating $\det(G_j + t\dot{G}_j) = \det G_j \det(I + tG_j^{-1}\dot{G}_j)$. Thus

$$\frac{d}{dt} \log \omega_j = \text{Tr}(P_j E), \quad \frac{d}{dt} \log V_j = \text{Tr}(Q_j E). \quad (\text{EW.31})$$

All these formulas use the original coordinates before the final cyclic trace.

Put $\mathcal{L} = \log \mathcal{R} \geq 0$ and $s = \min(q, r)$. The endpoint derivatives are exactly

$$\begin{aligned} (\log U)' &= \text{Tr}((P_m - P_n)E), \\ \mathcal{L}' &= \text{Tr}((Q_{n-1} + Q_n - Q_{m-1} - Q_m)E), \\ \mathcal{E}'_{n,r} &= \frac{1}{2r} \left\{ \mathcal{E}_{n,r} (\log U)' + U^{1/(2r)} \cosh(\mathcal{L}/(2r)) \mathcal{L}' \right\}. \end{aligned} \quad (\text{EW.32})$$

These equalities also hold at $\mathcal{R} = 1$. They identify the exact paired sensitivity rather than separate errors of large source and relation determinants.

Theorem 5.1 (Intrinsic finite sensitivity). *Throughout every such positive continuously differentiable family,*

$$\boxed{|\log U'| \leq \mathfrak{o}_H(E), \quad |\mathcal{L}'| \leq \mathfrak{o}_H(E) \min\{2s, \sqrt{2s\mathcal{L}}\}.} \quad (\text{EW.33})$$

For any two positive source matrices $H^{(0)}, H^{(1)}$ in the same original coordinates, let

$$\kappa = \frac{\lambda_{\max}((H^{(0)})^{-1/2} H^{(1)} (H^{(0)})^{-1/2})}{\lambda_{\min}((H^{(0)})^{-1/2} H^{(1)} (H^{(0)})^{-1/2})} \geq 1. \quad (\text{EW.34})$$

Their original window values satisfy

$$\boxed{|\log(U^{(1)}/U^{(0)})| \leq \log \kappa, \quad |\sqrt{\mathcal{L}^{(1)}} - \sqrt{\mathcal{L}^{(0)}}| \leq \sqrt{s/2} \log \kappa.} \quad (\text{EW.35})$$

In particular these sensitivity bounds contain no independent source-size determinant exponent. They vanish for any scalar change $H^{(1)} = tH^{(0)}$, $t > 0$, exactly as the original endpoint does.

Proof. Let $\Lambda = H_m^{-1/2} E H_m^{-1/2}$. Subtracting $\frac{1}{2}(\lambda_{\max} \Lambda + \lambda_{\min} \Lambda)I$ leaves operator norm $\mathfrak{o}_H(E)/2$. This scalar subtraction does not change either trace in (EW.32): the two monic-line projections each have trace one and the four least-lift projections each have trace q , with the displayed signs summing to zero.

The two monic orthogonal lines at degrees $n < m$ are orthogonal in the common ambient source. Their projection difference has eigenvalues 1, -1 and zeros, hence trace norm two. For any self-adjoint A and B , diagonalizing A gives $|\text{Tr}(AB)| \leq \sum_i |\lambda_i(A)| \|B\|$. This proves the first bound in (EW.33).

For the second trace split its four projections as $(\Pi_{n-1} - \Pi_{m-1}) + (\Pi_n - \Pi_m)$. Both index gaps are exactly r . Put $\ell_1 = \log(V_{n-1}/V_{m-1})$ and $\ell_2 = \log(V_n/V_m)$, so $\ell_1 + \ell_2 = \mathcal{L}$. Equation (EW.28), the triangle inequality for the trace norm, and $\sqrt{\ell_1} + \sqrt{\ell_2} \leq \sqrt{2\mathcal{L}}$ give

$$\|\Pi_{n-1} + \Pi_n - \Pi_{m-1} - \Pi_m\|_1 \leq 2\sqrt{2s\mathcal{L}}, \quad \|\Pi_{n-1} + \Pi_n - \Pi_{m-1} - \Pi_m\|_1 \leq 4s.$$

Multiplying either bound by $\mathfrak{o}_H(E)/2$ proves the second bound of (EW.33).

For a continuously differentiable path and $\varepsilon > 0$, the exact bound $|\mathcal{L}'| \leq \mathfrak{o}_H(E)\sqrt{2s\mathcal{L}}$ implies

$$\left| \frac{d}{dt} \sqrt{\mathcal{L} + \varepsilon} \right| \leq \sqrt{s/2} \mathfrak{o}_H(E).$$

Integrate on the compact path and let ε decrease to zero. This proves the square-root Lipschitz assertion along a path, including points at which $\mathcal{L} = 0$; no division by zero is used.

Use the straight positive segment $H(t) = (1-t)H^{(0)} + tH^{(1)}$, $0 \leq t \leq 1$. Let $\ell_i > 0$ be the eigenvalues of $(H^{(0)})^{-1/2}H^{(1)}(H^{(0)})^{-1/2}$. Congruence by $(H^{(0)})^{1/2}$ and then its displayed orthonormal eigenbasis gives the generalized eigenvalues of $(E, H(t))$ exactly as

$$\frac{\ell_i - 1}{1 + t(\ell_i - 1)}.$$

These are the eigenvalues of $H(t)^{-1/2}EH(t)^{-1/2}$: both follow from the same equation $Ex = \lambda H(t)x$. The displayed function is increasing in ℓ_i , with derivative $[1 + t(\ell_i - 1)]^{-2}$. Hence

$$\int_0^1 \mathfrak{o}_{H(t)}(E) dt = \log \ell_{\max} - \log \ell_{\min} = \log \kappa.$$

Integrating the already proved differential inequalities proves (EW.35). Finally direct substitution in (EW.11) gives $R_N(tH) = R_N(H)$, $G_N(tH) = tG_N(H)$, and $\omega_j(tH) = t\omega_j(H)$. Thus U and $\mathcal{R}$ are exactly unchanged by that explicitly stated common scaling. The source mass remains $t\mu_h^k$ in the scaled comparison; it was never reassigned the value one. $\square$

The independent projection review supplied a stronger exact version from the same angles. For $s \geq 1$, it is

$$|\mathcal{L}'| \leq 2s \mathfrak{o}_H(E) \sqrt{1 - e^{-\mathcal{L}/(2s)}}. \quad (\text{EW.35a})$$

To prove it, list $x_i = -\log \sigma_i^2$ for each of the two projection pairs used above, omitting their zero terms and then appending zeros to make exactly $2s$ entries. Their sum is $\mathcal{L}$. The exact trace-norm formula and the scalar cancellation prove $|\mathcal{L}'| \leq \mathfrak{o}_H(E) \sum_i f(x_i)$, where $f(x) = \sqrt{1 - e^{-x}}$. For $x > 0$,

$$f''(x) = -\frac{e^{-x}}{2\sqrt{1 - e^{-x}}} - \frac{e^{-2x}}{4(1 - e^{-x})^{3/2}} < 0.$$

Continuity at zero extends concavity there: apply the concavity inequality to positive arguments increased by ε and then let $\varepsilon \downarrow 0$. Finite Jensen on the full $2s$ -term list proves (EW.35a). Since $1 - e^{-x} \leq x$ and $1 - e^{-x} \leq 1$, it implies both bounds on $|\mathcal{L}'|$ in (EW.33).

Define the explicit increasing map on the actual loss variable by

$$\Phi_s : [0, \infty) \longrightarrow [0, \infty), \quad \Phi_s(L) = 2 \operatorname{arcosh}(e^{L/(4s)}), \quad \Phi_s^{-1}(v) = 4s \log \cosh(v/2).$$

Here $\operatorname{arcosh} x = \log(x + \sqrt{x^2 - 1})$ for $x \geq 1$, so both maps and their inverses have specified real branches. For $L > 0$, $\Phi'_s(L) = [2s\sqrt{1 - e^{-L/(2s)}}]^{-1}$. Apply (EW.35a) to $\Phi_s(\mathcal{L} + \varepsilon)$. Its derivative in absolute value is at most $\mathfrak{o}_H(E) \sqrt{(1 - e^{-\mathcal{L}/(2s)})/(1 - e^{-(\mathcal{L} + \varepsilon)/(2s)})} \leq \mathfrak{o}_H(E)$. Integrating and letting $\varepsilon \downarrow 0$, then using the exact straight-segment length already proved, gives

$$|\Phi_s(\mathcal{L}^{(1)}) - \Phi_s(\mathcal{L}^{(0)})| \leq \log \kappa. \quad (\text{EW.35b})$$

This conclusion includes zero losses by continuity. For $s = 0$, the only loss is zero and define the one-point map $\Phi_0(0) = 0$; the projection and endpoint comparisons are their zero identities.

For $s \geq 1$, the stronger enclosure itself has an exact algebraic form. Given any $\rho \geq \kappa$ and the original $\mathcal{R}^{(0)} \geq 1$, put

$$z_0 = (\mathcal{R}^{(0)})^{1/(4s)} + \sqrt{(\mathcal{R}^{(0)})^{1/(2s)} - 1}, \quad z_- = \max\{1, z_0/\sqrt{\rho}\}, \quad z_+ = \sqrt{\rho} z_0, \quad (\text{EW.35c})$$

$$\left(\frac{z_- + z_-^{-1}}{2} \right)^{4s} \leq \mathcal{R}^{(1)} \leq \left(\frac{z_+ + z_+^{-1}}{2} \right)^{4s}.$$

Indeed $z_0 = \exp(\Phi_s(\mathcal{L}^{(0)})/2)$, (EW.35b) encloses its counterpart between z_- and z_+ , and $z \mapsto (z + z^{-1})/2$ is increasing for $z \geq 1$. The identity $\mathcal{R} = \cosh(\Phi_s(\log \mathcal{R})/2)^{4s}$ proves (EW.35c). At the same time $U^{(0)}/\rho \leq U^{(1)} \leq \rho U^{(0)}$ by (EW.35). Thus monotonicity in (EW.20) supplies an endpoint enclosure using only nonnegative real roots, rational operations and integer powers when the central U , $\mathcal{R}$ and a bound ρ are rational. For example any proved rational Loewner comparison $aH^{(0)} \preceq H^{(1)} \preceq bH^{(0)}$, $0 < a \leq b$, gives $\rho = b/a$: the extreme Rayleigh quotients lie between a and b . This is the exact map from a positive-matrix certificate to the sharper window certificate, with the original mass and relation unchanged.

The square-root assertion is a Lipschitz assertion, and does not assert ordinary differentiability at every zero of $\mathcal{L}$. For an exact auxiliary illustration take $q = 1$, $\chi = S$, $n = 1$, $r = 1$, and the positive original-coordinate matrix

$$H_2(t) = \begin{pmatrix} 1 & 0 & t \\ 0 & 1 & 0 \\ t & 0 & 1 \end{pmatrix}, \quad |t| < 1.$$

Its leading pivots are $1, 1, 1 - t^2$, its remainder is evaluation at $S = 0$, and (EW.11) gives $V_0 = V_1 = 1$, $V_2 = 1 - t^2$. Thus $\mathcal{L} = -\log(1 - t^2)$ and $\sqrt{\mathcal{L}}/|t| \rightarrow 1$ as $t \rightarrow 0$. Its square root has a corner. This positive-form example belongs exactly to the finite family domain of (EW.33)–(EW.35); it is not claimed to be an arithmetic moment source. The regularized proof above includes it.

For completeness, (EW.35) has a directly usable endpoint enclosure. Writing $d_H = \log \kappa$ and $b = \sqrt{s/2} d_H$, it proves

$$\begin{aligned} e^{-d_H} U^{(0)} &\leq U^{(1)} \leq e^{d_H} U^{(0)}, \\ (\max\{0, \sqrt{\mathcal{L}^{(0)}} - b\})^2 &\leq \mathcal{L}^{(1)} \leq (\sqrt{\mathcal{L}^{(0)}} + b)^2. \end{aligned} \tag{EW.36}$$

The function $U^{1/(2r)} \sinh(\mathcal{L}/(2r))$ is increasing in each positive U and nonnegative $\mathcal{L}$, as direct differentiation shows. Substitution of the two endpoints in (EW.36) therefore bounds the actual endpoint expression. This is a finite stability theorem, including its exact $s = \min(q, r)$ dependence.

6 Rational interval certificates for the original input

Here is an explicit finite algorithm which uses enclosures of the original coefficients; it does not replace them by an arbitrary moment matrix. For clarity first suppose the constants in (EW.5) and the polynomial tests in (EW.10) are exact. Given real intervals $|c_{h,j} - \hat{c}_j| \leq e_j$ for $0 \leq j \leq 2m$, set

$$\begin{aligned} \hat{A}(z) &= \sum_{j=0}^{2m} (\alpha)_j \hat{c}_j z^j, \\ A_+(z) &= \sum_{j=0}^{2m} (\alpha)_j |\hat{c}_j| z^j, \quad E_+(z) = \sum_{j=0}^{2m} (\alpha)_j e_j z^j, \\ \hat{d}_l &= [z^l] \hat{A}(z)^k, \quad e_l^{(k)} = [z^l] \{(A_+(z) + E_+(z))^k - A_+(z)^k\}. \end{aligned} \tag{EW.37}$$

The multinomial expansion groups each difference of products by the nonempty set of factors carrying an error. Taking absolute values term by term proves $|d_l - \hat{d}_l| \leq e_l^{(k)}$. Thus the Hermitian central matrix and entry radii

$$\hat{H}_{ab} = C_k \sum_l l! L_{ab,l} \hat{d}_l, \quad \varepsilon_{ab} = C_k \sum_l l! |L_{ab,l}| e_l^{(k)} \tag{EW.38}$$

enclose the actual source. Rational upper bounds for the moduli give rational radii. An uncertain real constant, including C_k , is propagated by interval multiplication before the final matrix is

reported. For $|x - \hat{x}| \leq e_x$ and $|y - \hat{y}| \leq e_y$, the exact elementary rule is $|xy - \hat{x}\hat{y}| \leq |\hat{x}|e_y + |\hat{y}|e_x + e_x e_y$. Repeated use handles every finite product in (EW.8)–(EW.10).

The original relation need not have rational coefficients. To retain it exactly while producing rational certificates, use its quotient-first coordinate map at each endpoint degree N :

$$A_N = [E_N, B_N] : \mathbb{C}^q \oplus \mathbb{C}^{N-q+1} \longrightarrow \mathbb{C}^{N+1}, \quad \det A_N = 1, \quad \tilde{H}_N = A_N^* H_N A_N. \quad (\text{EW.39})$$

The first component of $A_N^{-1}P$ is exactly $J_N P$ by monic division. Supply rational enclosures of the real and imaginary parts of the actual χ coefficients when they are not exact rational numbers. Ordinary complex interval addition, multiplication and conjugation in (EW.39) give rational entry enclosures of the actual $\tilde{H}_N$. For a complex interval a modulus radius is bounded rationally, for example by the sum of its two coordinate radii. Conjugation changes the sign of the imaginary coordinate; it is applied to the actual first factor. This procedure never changes χ , its degree, its monicity or the original quotient coordinates. Intervals for λ and its positive constants, if required, are propagated in the same finite operations and in the specified recurrence (EW.6).

Here is a rational matrix certificate valid for either H_m or any of these $\tilde{H}_N$. Choose a Hermitian rational centre $\widehat{M}$ and symmetric rational radii ε_{ab} enclosing its actual error. Choose any positive rational v_a and form

$$\Delta = \text{diag} \left(\sum_b \varepsilon_{ab} \frac{v_b}{v_a} \right), \quad M^- = \widehat{M} - \Delta, \quad M^+ = \widehat{M} + \Delta. \quad (\text{EW.40})$$

The pairwise inequality $2|x_a||x_b| \leq (v_b/v_a)|x_a|^2 + (v_a/v_b)|x_b|^2$ proves $M^- \preceq M \preceq M^+$. Rational complex arithmetic now eliminates each leading pivot by the congruence

$$\begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \longmapsto \text{diag}(a, C - bb^*/a), \quad (x, y) \longmapsto (x - a^{-1}b^*y, y). \quad (\text{EW.41})$$

Recording strictly positive rational pivots proves $M^- > 0$ by induction and hence also $M^+ > 0$. A nonpositive pivot rejects that particular proposed certificate. Every actual matrix is already positive by its source integral and, for (EW.39), its invertible congruence. At a fixed finite degree, convergent valid entry intervals eventually give positive lower matrices: the least unit-sphere value of the actual quadratic form is positive by compactness, and the finite sum of entry errors tends to zero. This proves the finite certificate's convergence without assuming or claiming that a growing-degree arithmetic input has already been enclosed.

For the two norm endpoints take the corresponding principal submatrices of the certified H_m^-, H_m^+ . Define ω_j^-, ω_j^+ by their exact monic Schur complements. The squared norm of any fixed monic polynomial is between its two enclosing forms. Minimizing over the same affine monic set proves

$$0 < \omega_j^- \leq \omega_j \leq \omega_j^+, \quad j = n, m. \quad (\text{EW.42})$$

For each of $N = n - 1, n, m - 1, m$, use (EW.39)–(EW.41) and split its first q and last $N - q + 1$ coordinates. The Schur complements of the latter block in $\tilde{H}_N^-, \tilde{H}_N^+$ give rational positive forms G_N^-, G_N^+ . Their affine fibre has fixed first coordinate x , by the exact map (EW.39). Minimization therefore proves

$$G_N^- \preceq G_N \preceq G_N^+, \quad 0 < V_N^- := \det G_N^- \leq V_N \leq \det G_N^+ =: V_N^+. \quad (\text{EW.43})$$

For the determinant inequality, conjugate by the square root of the smaller positive form and multiply the resulting eigenvalues, each at least one. At $N = q - 1$ the boundary block is empty and the Schur complement is the whole q -dimensional form.

All numbers in the following enclosure are rational and positive:

$$\begin{aligned} U_- &= \omega_m^- / \omega_n^+, & U_+ &= \omega_m^+ / \omega_n^-, \\ \mathcal{R}_- &= \max \left\{ 1, \frac{V_{n-1}^- V_n^-}{V_{m-1}^+ V_m^+} \right\}, & \mathcal{R}_+ &= \frac{V_{n-1}^+ V_n^+}{V_{m-1}^- V_m^-}. \end{aligned} \quad (\text{EW.44})$$

They satisfy $U_- \leq U \leq U_+$ and $1 \leq \mathcal{R}_- \leq \mathcal{R} \leq \mathcal{R}_+$. The use of the lower bound one follows from the proved nested minimization, rather than an interval rounding choice.

Positive roots require no floating logarithm certificate. For any positive rational x and integer $2r$, find rational numbers $l, u > 0$ such that $l^{2r} \leq x \leq u^{2r}$. Doubling upper or lower powers of two supplies an initial interval, and ordinary rational bisection makes it arbitrarily narrow. Each decision is a positive-denominator integer comparison. Strict monotonicity of $t \mapsto t^{2r}$ for $t > 0$ proves the enclosure and convergence. Apply this algorithm to

$$U_- \mathcal{R}_-, \quad U_- / \mathcal{R}_-, \quad U_+ \mathcal{R}_+, \quad U_+ / \mathcal{R}_+.$$

Write their certified root intervals as $[l_1, u_1], \dots, [l_4, u_4]$. The function in (EW.20) is increasing in $U > 0$ and $\mathcal{R} \geq 1$: its U derivative is its nonnegative value times $1/(2rU)$, and its $\mathcal{R}$ derivative is $U^{1/(2r)} \cosh(\log \mathcal{R}/(2r)) / (2r\mathcal{R}) > 0$. Therefore the complete rational endpoint certificate is

$$\boxed{\max\{0, (l_1 - u_2)/2\} \leq \mathcal{E}_{n,r} \leq (u_3 - l_4)/2.} \quad (\text{EW.45})$$

Alternatively (EW.22) gives a positive numerator and denominator certificate from the same root brackets. Every matrix pivot, integer power comparison, relation coefficient and source mass used in this algorithm is inspectable. It makes no claim for coefficient intervals which have not been provided.

7 Boundary cases and the retained cochain map

At $r = 1$, the block has two columns, $W_r = w$, and $\mathcal{R} = \mathcal{R}_2$ because the two copies of $\mathcal{R}_1$ cancel. Equations (EW.18)–(EW.22) reduce exactly to the two-degree formula GMT.24–27. At $n = q$, the base source degree is $q - 1$ and its boundary dimension is zero; its quotient metric is simply the source Gram. All formulas above still use positive matrices of their specified sizes. No degree $q - 2$ full-quotient form is inserted.

The actual positive polynomial source has an additional strictness property for $q \geq 1$: every two-degree quotient-volume ratio is strictly greater than one. Here is the complete algebraic reason. Let $\pi_j(u)$ be the real monic orthogonal family of the original positive real- u measure. Multiplication by u is self-adjoint. Pairing $u\pi_j$ with lower orthogonal polynomials and using their degrees gives

$$u\pi_j = \pi_{j+1} + b_j\pi_j + a_j\pi_{j-1}, \quad b_j \in \mathbb{R}, \quad a_j = \omega_j/\omega_{j-1} > 0 \quad (j \geq 1).$$

For $j = 0$ the last term is absent. The exact monic phase map $p_j(S) = i^j \pi_j((S - c)/i)$ has the same squared norm, because $|i^j| = 1$, and gives

$$Sp_j = p_{j+1} + (c + ib_j)p_j - a_j p_{j-1}.$$

If a nonconstant χ divided p_j and p_{j+1} , this recurrence and $a_j \neq 0$ would make it divide p_{j-1} . Induction would make it divide $p_0 = 1$, a contradiction. For the two appended columns at degrees $N, N + 1$ over $\mathcal{P}_{N-1}$, the residuals are p_N and $p_{N+1} + zp_N$, where the original coefficient is exactly $z = \langle p_N, S^{N+1} \rangle_H / \omega_N$. Indeed their difference from those polynomials has smaller degree, and both residuals are orthogonal to $\mathcal{P}_{N-1}$. If their full quotient residual map F_2 were zero, both p_N and p_{N+1} would have zero remainder, contradicting the divisibility argument. Thus $Z_2 = F_2^* G_{N-1} F_2$ is a nonzero positive semidefinite matrix, and (EW.17) gives $V_{N-1}/V_{N+1} > 1$ for every $N \geq q$. The finite product

$$\mathcal{R} = \prod_{j=0}^{r-1} \frac{V_{n+j-1}}{V_{n+j+1}} > 1 \quad (q \geq 1, n \geq q, r \geq 1) \quad (\text{EW.46a})$$

is its exact overlapping telescope. This proves strict positivity of the actual $\mathcal{E}_{n,r}$ and of $\mathcal{L}$ at each admitted nonempty arithmetic window; it supplies no uniform lower bound as degrees or

packets grow. The more general positive-form sensitivity theorem includes the zero-loss boundary separately.

The empty packet is a separate declared case: $h = 1$, $\chi = 1$, $q = 0$, and $\mathcal{C} = 0$. The analytic density and mass remain those of (EW.1). Put $V_N = 1$ even at the empty degree $N = -1$; all quotient matrices, projections and F_j have their empty dimensions. For $n \geq 0, r \geq 1$ the source norms exist, $\mathcal{R} = 1$, and $\mathcal{E}_{n,r} = 0$. If $n = 0$, use the empty base source H_{-1} , with determinant one and no inverse operation; the block $W = H_r$ then has its original monomial coordinates. Its first entry is the full mass. The projection and volume sensitivity statements become their zero-map identities, while the source-norm sensitivity remains the proved rank-two calculation.

Finally the projection comparison (EW.26) has an explicit original theta representative. For $a < b$ and $x \in \mathcal{C}$, monic division gives a unique polynomial $\kappa_x \in \mathcal{P}_{b-q}$ with

$$\iota_a R_a x - \iota_b R_b x = \chi \kappa_x. \quad (\text{EW.46})$$

Its coefficients are obtained by the same exact division, so the map $x \mapsto \kappa_x$ is linear. Let $\mathcal{V}_{h,k} P = P(D_1 + \dots + D_k) F_h^{\otimes k}$, where $D = -x \partial_x$, $\mathcal{M} F_h = v_h$, and $h(D) F_h = \Theta \phi_0$. Successive division in $s_1, \dots, s_k$ supplies specific Q_i with $\chi(\sum s_i) = \sum_i h(s_i) Q_i(\mathbf{s})$. The final tensor remainder vanishes because its monomials form a basis of $(\mathbb{C}[s]/(h))^{\otimes k}$ and χ annihilates the original sum. Thus the exact cochain of degree $k - 1$

$$\mathcal{P}_\chi(\kappa_x) = \sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) \kappa_x(\sum D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}) \quad (\text{EW.47})$$

satisfies

$$d_{\text{tot}} \mathcal{P}_\chi(\kappa_x) = \mathcal{V}_{h,k} (\iota_a R_a x - \iota_b R_b x). \quad (\text{EW.48})$$

Indeed the tensor differential at position i contributes $(-1)^{i-1}$ because its preceding factors have cochain degree one; this cancels the written sign, and $h(D_i) F_h = \Theta \phi_0$ then proves the equality term by term. The arithmetic jet of both representatives is the same ηx of (EW.3). In the original support carrier $\{\tau\} \sqcup \bigsqcup_\ell (\{\ell\} \times E_\ell)$, a fibrewise linear map sends (ℓ, v) to $(\ell, L_\ell v)$ and τ to τ . Thus the displayed boundary maps to $e_\ell = (\ell, 0)$ while external absence remains τ . The endpoint's representative changes, projection angles, and volume changes have therefore all been related to the original typed maps and their full theta primitives.

8 Verification scope and exact auxiliary calibration

The companion checker uses the explicitly auxiliary weight

$$\begin{aligned} \lambda = 1, \quad k = 2, \quad c = 1, \quad c_1 = 1/2, \quad C_2 = 1/4, \\ B(t) = 1 + t^2/4, \quad w = Br_1, \quad \chi(S) = S^2 - 2S - 1. \end{aligned} \quad (\text{EW.49})$$

This is a specified positive $L^2(r_1)$ multiplier and the same finite coefficient-to-Gram domain as (EW.8)–(EW.10). It is not asserted to be an actual zeta packet. Its exact connection to the arithmetic problem is the complete sequence of maps (EW.10), (EW.11), (EW.14)–(EW.23), and (EW.39)–(EW.45), with its auxiliary inputs written explicitly.

For this fixture $b_2^{(1)} = t^2 - 2$, so $c_{B,0} = 3/2$, $c_{B,2} = 1/4$, and every other coefficient is zero. Consequently $\sum_j (2)_j c_{B,j} z^j = 3/2 + 3z^2/2$ and its square has coefficients $d_0 = 9/4$, $d_2 = 9/2$, $d_4 = 9/4$. Its one-factor mass is $3/4$, the literal sum mass is $9/16$, and the reference sum mass is $1/4$. The checker independently constructs Gamma moments from the monic recurrence, inserts $1 + t^2/4$ in each factor, and expands every power of $t_1 + t_2$ with its full binomial coefficient. It also constructs the coefficient-side moments from (EW.9); agreement compares two exact finite calculations, including the tensor cross terms.

In this auxiliary example gamma recurrence gives a further exact all-degree check. Since $|\Gamma(2 + it/2)|^2 = (1 + t^2/4)|\Gamma(1 + it/2)|^2$, one has $Br_1 = r_2$. The displayed masses give $c_2 = 3/4$, $c_4 = 315/8$, and hence

$$m = r_2 * r_2 = r_4/70, \quad r_1 * r_1 = r_2/3, \quad \omega_j = \frac{9}{16}j!(8)_j, \quad U = (n+1)_r(n+8)_r. \quad (\text{EW.50})$$

The first two equalities follow from the literal convolution law (EW.7), with coefficients $(3/4)^2/(315/8) = 1/70$ and $(1/2)^2/(3/4) = 1/3$. The monic original-coordinate map is $p_j(S) = i^j b_j^{(4)}((S-1)/i)$, whose squared norm is exactly the third quantity by (EW.7). Taking the stated endpoint ratio proves the final formula.

The exact selected windows computed in the accompanying rational receipt are

(n, r)	U	$\mathcal{R}$
$(2, 1)$	30	20191/4860
$(2, 2)$	1320	3073295611/216513000
$(3, 2)$	2640	295913127219469/36104825790000

(EW.51)

Here is a full independent finite derivation of the entries in that table, using the original quotient kernel rather than a Schur block. Let $x = S - 1$ solely in the stated quotient-coordinate map

$$A : \mathbb{C}_{(1,x)}^2 \longrightarrow \mathbb{C}_{(1,S)}^2, \quad (a, b) \longmapsto (a - b, b), \quad A = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad \det A = 1.$$

It takes the identical relation to $x^2 - 2$, so its residue equation is exactly $x^2 = 2$. It preserves the determinant of the quotient form by the congruence $G_N^x = A^* G_N A$. The monic source family in (EW.50) obeys $p_{j+1} = xp_j + j(j+7)p_{j-1}$, by applying the exact phase map to (EW.6). Orthogonal expansion of the original source inverse in (EW.11) proves the finite quotient kernel formula

$$(G_N^x)^{-1} = \sum_{j=0}^N \frac{[p_j]_{(1,x)} [p_j]_{(1,x)}^*}{\omega_j}. \quad (\text{EW.52})$$

Indeed the matrix of the monic orthogonal columns in the original source is invertible triangular with diagonal one, takes H_N to $\text{diag}(\omega_0, \dots, \omega_N)$, and sends the reduction columns to their written remainders. Substituting that inverse congruence into $J_N H_N^{-1} J_N^*$ gives (EW.52) term by term. The recurrence and $x^2 = 2$ give the complete required list

j	$[p_j]_{(1,x)}$	ω_j
0	(1, 0)	9/16
1	(0, 1)	9/2
2	(10, 0)	81
3	(0, 28)	2430
4	(356, 0)	106920
5	(0, 1588)	6415200

(EW.53)

For example the successive remainders are obtained from $x^2 + 8 = 10$, $10x + 18x = 28x$, $28x^2 + 300 = 356$, and $356x + 44 \cdot 28x = 1588x$. The positive kernels from (EW.52) are diagonal in this explicitly transported quotient basis, with

N	$((G_N^x)^{-1})_{00}$	$((G_N^x)^{-1})_{11}$	V_N
1	16/9	2/9	81/32
2	244/81	2/9	729/488
3	244/81	662/1215	98415/161528
4	56102/13365	662/1215	16238475/37139524
5	56102/13365	376069/400950	2679348375/10549111519

(EW.54)

Each diagonal entry is the indicated finite sum of rational terms in (EW.52), and V_N is the reciprocal of their product because $\det A = 1$. Substituting (EW.53)–(EW.54) in (EW.18) gives all three rows of (EW.51) by positive-denominator multiplication.

There is also an executed coefficient-box certificate, before any matrix error is chosen. Put $\varepsilon = 10^{-12}$ and retain $n = 2, r = 1, m = 3$, the same reference mass and the same χ . Allow the complete specified one-factor input box

$$|c_{B,0} - 3/2| \leq \varepsilon, \quad |c_{B,2} - 1/4| \leq \varepsilon, \quad c_{B,j} = 0 \quad (j = 1, 3, 4, 5, 6). \quad (\text{EW.55})$$

Every point of this box is realized by the specified positive multiplier $B_{a,b}(t) = a + bt^2$, where $a = c_{B,0} - 2c_{B,2}$ and $b = c_{B,2}$. They obey $a \geq 1 - 3\varepsilon > 0$ and $b \geq 1/4 - \varepsilon > 0$. Its coefficient formula follows by $a + bt^2 = (a + 2b)b_0^{(1)} + bb_2^{(1)}$. The interior input $a = 1 + \varepsilon$, $b = 1/4 - \varepsilon/4$ has one-factor mass $3/4 + \varepsilon/4$ and sum mass $(3/4 + \varepsilon/4)^2$. Thus neither the box nor its test input replaces the zeroth coefficient by a unit mass.

For (EW.55), the complete coefficient-power error formula (EW.37) is exactly

$$\begin{aligned} e_0^{(2)} &= 3\varepsilon + \varepsilon^2, & e_2^{(2)} &= 21\varepsilon + 12\varepsilon^2, & e_4^{(2)} &= 18\varepsilon + 36\varepsilon^2, \\ e_1^{(2)} &= e_3^{(2)} = e_5^{(2)} = e_6^{(2)} = 0. \end{aligned} \quad (\text{EW.56})$$

Indeed the central polynomial is $3/2 + (3/2)z^2$ and its error polynomial is $\varepsilon + 6\varepsilon z^2$; expanding the full square difference gives the three displayed coefficients. The upper corner of the coefficient box attains all three positive differences, so its quadratic error terms cannot be omitted.

Expanding all sixteen original tests $(1 - iu)^a(1 + iu)^b$, $0 \leq a, b \leq 3$, in the monic Gamma basis of (EW.6), then using (EW.38)–(EW.40) with four weights equal to one, gives

$$\begin{aligned} \hat{H}_3 &= \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 \\ 9 & 81 & 81 & -1863 \\ -63 & 81 & 2025 & 2025 \\ -207 & -1863 & 2025 & 93393 \end{pmatrix}, \\ \Delta_{00} &= 54\varepsilon + 28\varepsilon^2, & \Delta_{11} &= (1029\varepsilon + 873\varepsilon^2)/2, \\ \Delta_{22} &= (2109\varepsilon + 1773\varepsilon^2)/2, & \Delta_{33} &= 29754\varepsilon + 32878\varepsilon^2. \end{aligned} \quad (\text{EW.57})$$

These diagonal sums retain the absolute value of every test coefficient. The real coefficients occur at even degrees and the imaginary coefficients at odd degrees; the latter pair exactly to zero against every member of the specified even coefficient box. The complete complex expansions, including those zero pairings, are recorded in the certificate.

Here is an elementary exact positive-definiteness certificate for the whole box. Rational multiplication verifies

$$\hat{H}_3^{-1} = \frac{1}{2430} \begin{pmatrix} 8019 & -1863 & 351 & -27 \\ -1863 & 1501 & -147 & 29 \\ 351 & -147 & 39 & -3 \\ -27 & 29 & -3 & 1 \end{pmatrix}, \quad \text{Tr } \hat{H}_3^{-1} = 956/243 < 4.$$

The central form is positive by its actual measure, or by the four original monic norms in (EW.53). Its inverse is positive, so each of its eigenvalues is at most its trace. Consequently $\hat{H}_3 \succ I_4/4$. Substitution of $\varepsilon = 10^{-12}$ into the four full polynomials (EW.57) gives $0 < \Delta_{jj} < 10^{-6}$ by integer comparison. Hence

$$\hat{H}_3 - \Delta \succ (1/4 - 10^{-6})I_4 \succ 0, \quad \hat{H}_3 - \Delta \preceq H_3 \preceq \hat{H}_3 + \Delta \quad (\text{EW.58})$$

for every actual multiplier in (EW.55). This proves the positive matrix enclosure before its inverses are evaluated. The accompanying receipt also records each exact rational elimination pivot separately.

Use the original J_N and χ at $N = 1, 2, 3$ in (EW.42)–(EW.45). The resulting rational interval for every endpoint value in this whole coefficient box is

$$\boxed{\frac{10247872526420120792621067}{2^{81}} \leq \mathcal{E}_{2,1} \leq \frac{10247872551311149445602506}{2^{81}}}. \quad (\text{EW.59})$$

The four endpoint arguments in (EW.45) are computed from the written rational matrices $\widehat{H}_3 \pm \Delta$ by their monic and quotient Schur formulas; the receipt gives every one of these exact rational values and its two squared-root comparisons. Thus (EW.59) is certified by integer powers and the minimum principle, with no rounding assertion about a transcendental logarithm. The interior positive multiplier and all four box corners are independently reconstructed by the literal one-factor moment recurrence and tensor binomial expansion, and pass the same original-coordinate entry bounds.

All original- S Gram entries are computed from $(1 - iu)^a(1 + iu)^b$. The checker’s window certificates include the exact reduction matrix, source norms, quotient metrics, the $r + 1$ Schur block and all its prefixes, both cancellations in (EW.21), positive rational matrix envelopes, and the integer power comparisons for (EW.45). The written proof establishes the all-degree statements; each executed finite check is reported separately in the accompanying receipt. The base fixture executes 416 distinct mathematical checks across fifteen windows; its normal and optimized runs pass and six changed-formula runs in each mode are rejected after executing the complete suite. The separate supplement executes 329 mathematical checks and two dependency-hash checks; its four changed-formula runs in each mode are also rejected after the complete suite, including the omitted quadratic coefficient error at an attained box corner. The independent remainder-kernel calibration executes the six scalar comparisons in (EW.51). These counts describe their separate scopes; repeated modes and changed-formula runs do not add new mathematical checks. Numerical examples do not stand in for the arithmetic input or a growing-window bound.